

هوش مصنوعی و کارگاه

دانشکده ریاضی و علوم کامپیوتر دانشگاه صنعتی امیرکبیر



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دانشجوی دکتری علوم کامپیوتر

Artificial Intelligence & Workshop

استنتاج، استنتاج احتمالاتی و کاربرد در
یادگیری ماشین

- Reasoning, Probabilistic Reasoning

Artificial Intelligence

<http://www.cs.duke.edu/courses/spring09/cps170/>

First-Order Logic

By: Professor Vincent Conitzer

Professor of CS, Carnegie Mellon University;
Professor of CS and Philosophy, University
of Oxford



منطق مرتبه اول و منطق مراتب بالاتر

منطق مرتبه اول و منطق مراتب بالاتر

- (First-Order Logic)

- منطق مرتبه اول: فقط روی متغیرهای فردی (اشیاء در دامنه بحث) کمیت‌سازی می‌کند.
(...X, Y)

- (Higher-Order Logic)

- منطق مراتب بالاتر: روی خواص، روابط و توابع (یعنی خود محمول‌ها و توابع) نیز می‌تواند کمیت‌سازی کند.

- مثال: اصل استقرا یک جمله در منطق مرتبه دوم است زیرا روی محمول P کمیت‌سازی شده است

$$\forall P ((P(0) \wedge \forall n (P(n) \rightarrow P(n+1))) \rightarrow \forall n P(n))$$

- یادگیری ماشین از کدام نوع منطق پشتیبانی می‌کند؟

Limitations of propositional logic (منطق گزاره‌ای)

- So far we studied propositional logic
- Some English statements are hard to model in propositional logic:
- “If your roommate is wet because of rain, your roommate must not be carrying **any** umbrella”
- Pathetic attempt at modeling this:
- $\text{RoommateWetBecauseOfRain} \Rightarrow (\text{NOT}(\text{RoommateCarryingUmbrella0}) \text{ AND } \text{NOT}(\text{RoommateCarryingUmbrella1}) \text{ AND } \text{NOT}(\text{RoommateCarryingUmbrella2}) \text{ AND } \dots)$

Problems with propositional logic

- No notion of objects
- No notion of relations among objects
- RoommateCarryingUmbrella0 is instructive **to us**, suggesting
 - there is an object we call Roommate,
 - there is an object we call Umbrella0,
 - there is a relationship Carrying between these two objects
- **Formally, none of this meaning is there**
 - Might as well have replaced RoommateCarryingUmbrella0 by P

Elements of first-order logic

- **Objects:** can give these names such as Umbrella0, Person0, John, Earth, ...
- **Relations:** Carrying(., .), IsAnUmbrella(.)
 - Carrying(Person0, Umbrella0), IsUmbrella(Umbrella0)
 - Relations with one object = **unary relations** = **properties**
- **Functions:** Roommate(.)
 - Roommate(Person0)
- **Equality:** Roommate(Person0) = Person1

Things to note about functions

- It could be that we have a separate name for Roommate(Person0)
 - E.g., Roommate(Person0) = Person1
 - ... but we do not **need** to have such a name
-
- A function can be applied to any object
 - E.g., Roommate(Umbrella0)

Reasoning about many objects at once

- **Variables:** $x, y, z, \dots$ can refer to multiple objects
- New operators “for all” and “there exists”
 - **Universal quantifier** and **existential quantifier**
- for all x : $\text{CompletelyWhite}(x) \Rightarrow \text{NOT}(\text{PartiallyBlack}(x))$
 - Completely white objects are never partially black
- there exists x : $\text{PartiallyWhite}(x) \text{ AND } \text{PartiallyBlack}(x)$
 - There exists some object in the world that is partially white and partially black

Practice converting English to first-order logic

- “John has Jane’s umbrella”
 - $\text{Has}(\text{John}, \text{Umbrella}(\text{Jane}))$
- “John has an umbrella”
 - there exists y : $(\text{Has}(\text{John}, y) \text{ AND } \text{IsUmbrella}(y))$
- “Anything that has an umbrella is not wet”
 - for all x : $((\text{there exists } y: (\text{Has}(x, y) \text{ AND } \text{IsUmbrella}(y))) \Rightarrow \text{NOT}(\text{IsWet}(x)))$
- “Any person who has an umbrella is not wet”
 - for all x : $(\text{IsPerson}(x) \Rightarrow ((\text{there exists } y: (\text{Has}(x, y) \text{ AND } \text{IsUmbrella}(y))) \Rightarrow \text{NOT}(\text{IsWet}(x))))$

Something we cannot do in first-order logic

- We are **not** allowed to reason in general about relations and functions
- The following would correspond to **higher-order logic** (which is more powerful):
 - “If John is Jack’s roommate, then any property of John is also a property of Jack’s roommate”
 - $(\text{John}=\text{Roommate}(\text{Jack})) \Rightarrow \text{for all } p: (p(\text{John}) \Rightarrow p(\text{Roommate}(\text{Jack})))$
- “If a property is inherited by children, then for any thing, if that property is true of it, it must also be true for any child of it”
- $\text{for all } p: (\text{IsInheritedByChildren}(p) \Rightarrow (\text{for all } x, y: ((\text{IsChildOf}(x,y) \text{ AND } p(y)) \Rightarrow p(x))))$

Generalized modus ponens

- for all x : Loves(John, x)
 - John loves every thing
- for all y : (Loves(y , Jane) $\Rightarrow$ FeelsAppreciatedBy(Jane, y))
 - Jane feels appreciated by every thing that loves her
- Can infer from this:
 - FeelsAppreciatedBy(Jane, John)
- Here, we used the substitution $\{x/\text{Jane}, y/\text{John}\}$
 - Note we used different variables for the different sentences
- General UNIFY algorithms for finding a good substitution

(Extremely informal statement of) Gödel's Incompleteness Theorem

- First-order logic is not rich enough to model basic arithmetic
- For any consistent system of axioms that is rich enough to capture basic arithmetic (in particular, mathematical induction), there exist true sentences that cannot be proved from those axioms
 - (قیاس دروغگو): فرض کنید یک سیستم (مثلاً یک ربات یا یک کتاب قانون) دارید که ادعا می کند "هر جمله ای که من بگویم، درست است".
 - حالا جمله مقابل را در نظر بگیرید: "این جمله، توسط این سیستم قابل اثبات نیست".
 - اگر فرض کنیم این جمله قابل اثبات است، یعنی سیستم می تواند ثابت کند که "من نمی توانم این جمله را اثبات کنم". این یک تناقض آشکار است! (چون اگر بتواند ثابتش کند، پس جمله کاذب است).
 - اگر فرض کنیم این جمله غیرقابل اثبات است، دقیقاً همان چیزی است که جمله ادعا می کند. پس جمله درست است، اما سیستم نمی تواند آن را اثبات کند

Artificial Intelligence

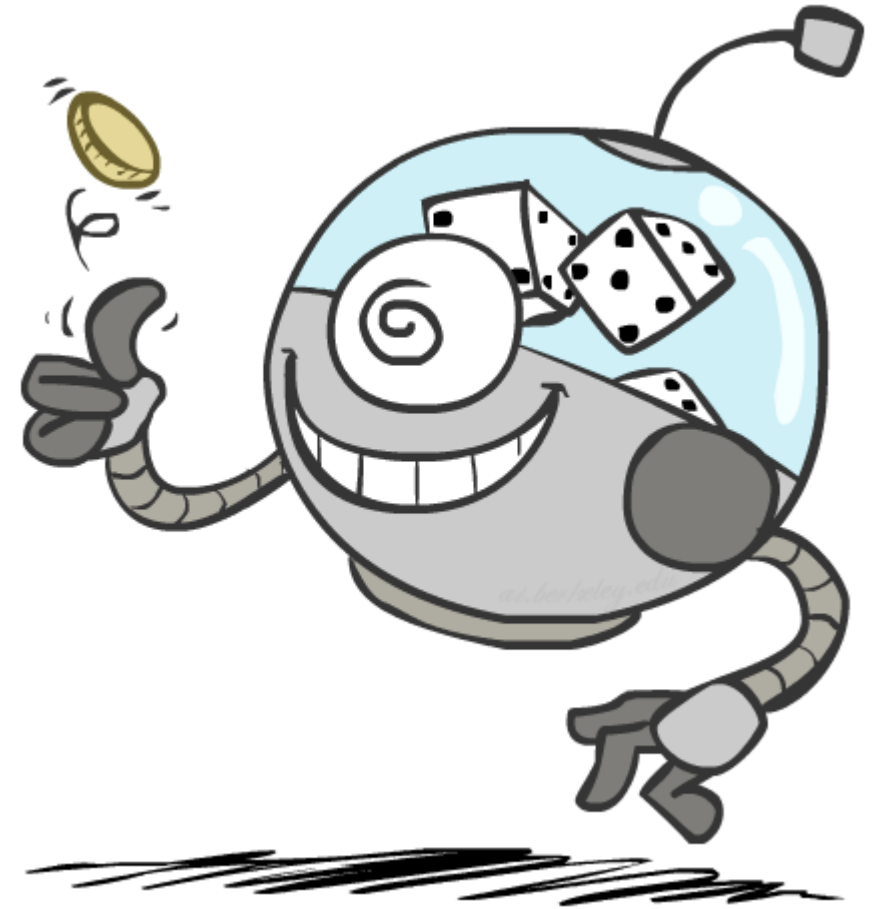
Probability & Probabilistic Reasoning



Instructor: Anca Dragan --- University of California, Berkeley

Today

- Probability
 - Random Variables
 - Joint and Marginal Distributions
 - Conditional Distribution
 - Product Rule, Chain Rule, Bayes' Rule
 - Inference
 - Independence



Uncertainty

- General situation:
 - **Observed variables (evidence):** Agent knows certain things about the state of the world (e.g., sensor readings or symptoms)
 - **Unobserved variables:** Agent needs to reason about other aspects (e.g. where an object is or what disease is present)
 - **Model:** Agent knows something about how the known variables relate to the unknown variables
- Probabilistic reasoning gives us a framework for managing our beliefs and knowledge

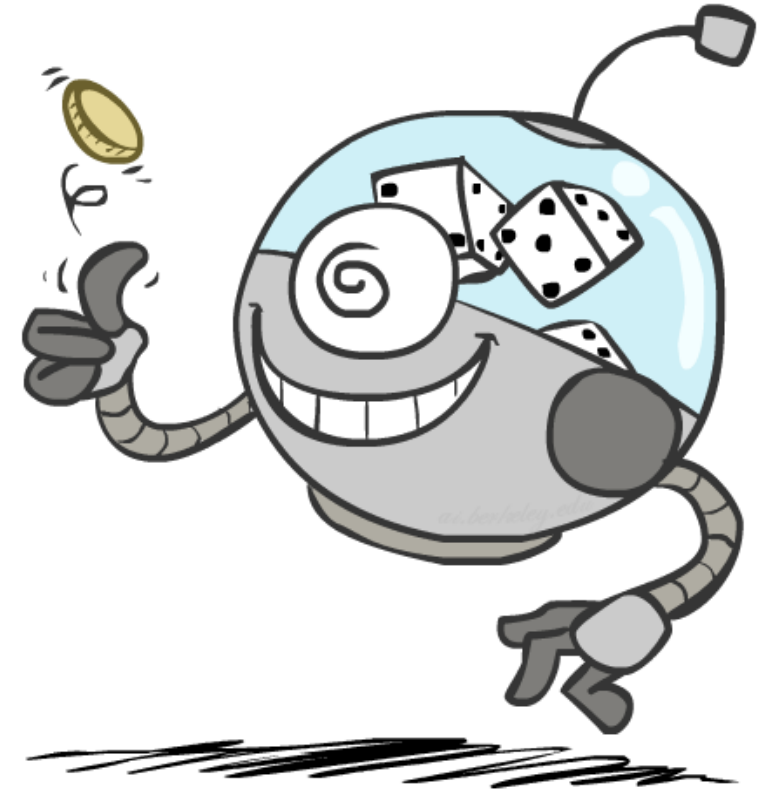
0.11	0.11	0.11
0.11	0.11	0.11
0.11	0.11	0.11

0.17	0.10	0.10
0.09	0.17	0.10
<0.01	0.09	0.17

<0.01	<0.01	0.03
<0.01	0.05	0.05
<0.01	0.05	0.81

Random Variables

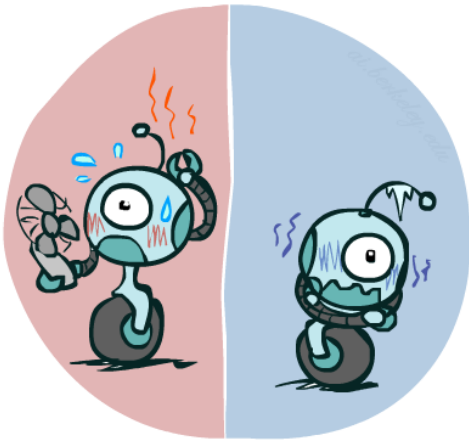
- A random variable is some aspect of the world about which we (may) have uncertainty
 - R = Is it raining?
 - T = Is it hot or cold?
 - D = How long will it take to drive to work?
 - L = Where is the ghost?
- We denote random variables with capital letters
- Like variables in a CSP, random variables have domains
 - R in $\{\text{true}, \text{false}\}$ (often write as $\{+r, -r\}$)
 - T in $\{\text{hot}, \text{cold}\}$
 - D in $[0, \infty)$
 - L in possible locations, maybe $\{(0,0), (0,1), \dots\}$



Probability Distributions

- Associate a probability with each value

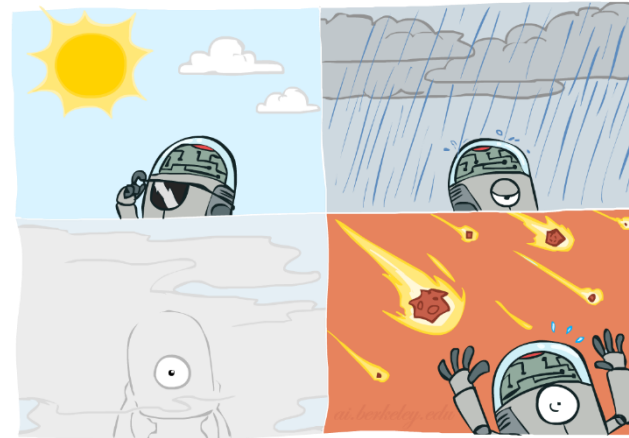
- Temperature:



$P(T)$

T	P
hot	0.5
cold	0.5

- Weather:



$P(W)$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

Probability Distributions

- Unobserved random variables have distributions

$P(T)$

T	P
hot	0.5
cold	0.5

$P(W)$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

Shorthand notation:

$$P(\textit{hot}) = P(T = \textit{hot}),$$

$$P(\textit{cold}) = P(T = \textit{cold}),$$

$$P(\textit{rain}) = P(W = \textit{rain}),$$

...

OK if all domain entries are unique

- A distribution is a TABLE of probabilities of values
- A probability (lower case value) is a single number

$$P(W = \textit{rain}) = 0.1$$

- Must have: $\forall x \ P(X = x) \geq 0$ and $\sum_x P(X = x) = 1$

Joint Distributions

- A *joint distribution* over a set of random variables: $X_1, X_2, \dots, X_n$ specifies a real number for each assignment (or *outcome*):

$$P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

$$P(x_1, x_2, \dots, x_n)$$

- Must obey: $P(x_1, x_2, \dots, x_n) \geq 0$

$$\sum_{(x_1, x_2, \dots, x_n)} P(x_1, x_2, \dots, x_n) = 1$$

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Probabilistic Models

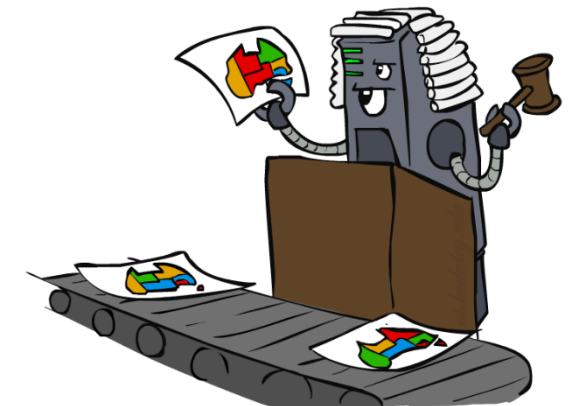
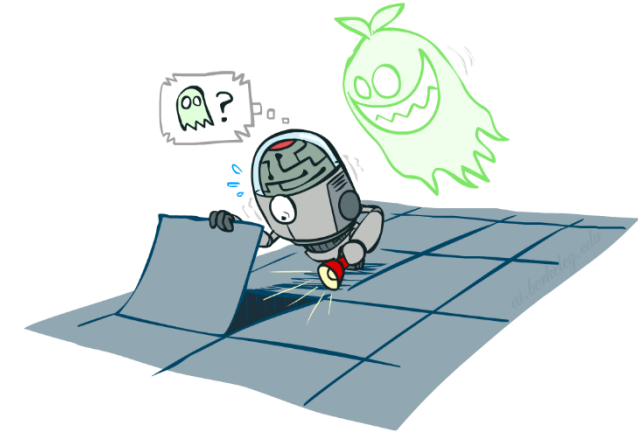
- A probabilistic model is a joint distribution over a set of random variables
- Probabilistic models:
 - (Random) variables with domains
 - Assignments are called *outcomes*
 - Joint distributions: say whether assignments (outcomes) are likely
 - *Normalized*: sum to 1.0
 - Ideally: only certain variables directly interact
- Constraint satisfaction problems:
 - Variables with domains
 - Constraints: state whether assignments are possible
 - Ideally: only certain variables directly interact

Distribution over T,W

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Constraint over T,W

T	W	P
hot	sun	T
hot	rain	F
cold	sun	F
cold	rain	T



Events

- An *event* is a set E of outcomes

$$P(E) = \sum_{(x_1 \dots x_n) \in E} P(x_1 \dots x_n)$$

- From a joint distribution, we can calculate the probability of any event
 - Probability that it's hot AND sunny?
 - Probability that it's hot?
 - Probability that it's hot OR sunny?
- Typically, the events we care about are *partial assignments*, like $P(T=\text{hot})$

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Quiz: Events

- $P(+x, +y)$?
- $P(+x)$?
- $P(-y \text{ OR } +x)$?

$P(X, Y)$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

Quiz: Events

- $P(+x, +y)$?

.2

- $P(+x)$?

.2+.3=.5

- $P(-y \text{ OR } +x)$?

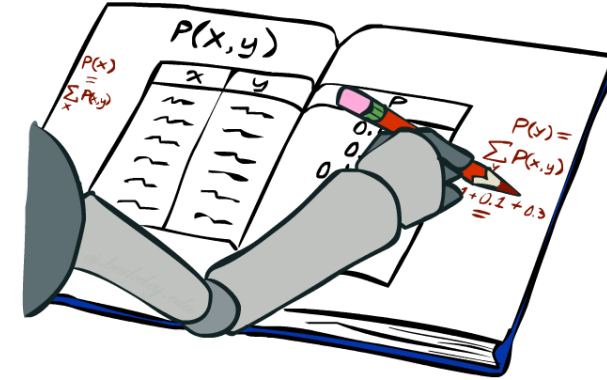
.1+.3+.2=.6

$P(X, Y)$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

Marginal Distributions

- Marginal distributions are sub-tables which eliminate variables
- Marginalization (summing out): Combine collapsed rows by adding



$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3



$$P(t) = \sum_s P(t, s)$$

$P(T)$

T	P
hot	0.5
cold	0.5



$$P(s) = \sum_t P(t, s)$$

$P(W)$

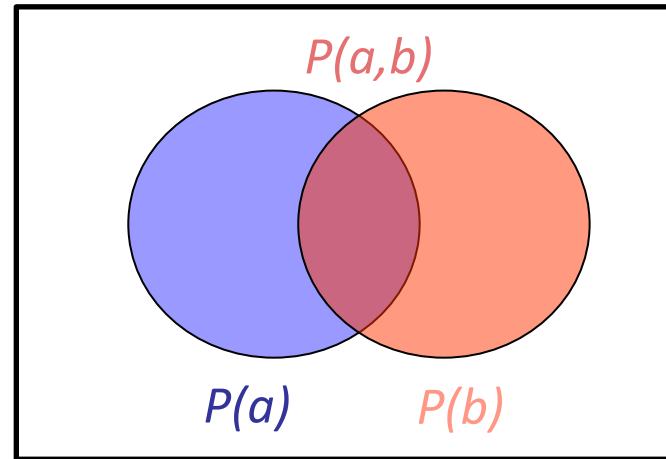
W	P
sun	0.6
rain	0.4

$$P(X_1 = x_1) = \sum_{x_2} P(X_1 = x_1, X_2 = x_2)$$

Conditional Probabilities

- A simple relation between joint and conditional probabilities
 - In fact, this is taken as the *definition* of a conditional probability

$$P(a|b) = \frac{P(a, b)}{P(b)}$$



$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(W = s|T = c) = \frac{P(W = s, T = c)}{P(T = c)} = \frac{0.2}{0.5} = 0.4$$

$$\begin{aligned} &= P(W = s, T = c) + P(W = r, T = c) \\ &= 0.2 + 0.3 = 0.5 \end{aligned}$$

Conditional Distributions

- Conditional distributions are probability distributions over some variables given fixed values of others

Conditional Distributions

$P(W T)$	$P(W T = \text{hot})$	
	W	P
	sun	0.8
	rain	0.2
	$P(W T = \text{cold})$	
	W	P
	sun	0.4
	rain	0.6

Joint Distribution

$P(T, W)$		
T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Normalization Trick

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$\begin{aligned}P(W = s|T = c) &= \frac{P(W = s, T = c)}{P(T = c)} \\&= \frac{P(W = s, T = c)}{P(W = s, T = c) + P(W = r, T = c)} \\&= \frac{0.2}{0.2 + 0.3} = 0.4\end{aligned}$$



$P(W|T = c)$

W	P
sun	0.4
rain	0.6

$$\begin{aligned}P(W = r|T = c) &= \frac{P(W = r, T = c)}{P(T = c)} \\&= \frac{P(W = r, T = c)}{P(W = s, T = c) + P(W = r, T = c)} \\&= \frac{0.3}{0.2 + 0.3} = 0.6\end{aligned}$$

Normalization Trick

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

SELECT the joint probabilities matching the evidence



$P(c, W)$

T	W	P
cold	sun	0.2
cold	rain	0.3

NORMALIZE the selection (make it sum to one)



$P(W|T = c)$

W	P
sun	0.4
rain	0.6

- Why does this work? Sum of selection is $P(\text{evidence})$! ($P(T=c)$, here)

$$P(x_1|x_2) = \frac{P(x_1, x_2)}{P(x_2)} = \frac{P(x_1, x_2)}{\sum_{x_1} P(x_1, x_2)}$$

To Normalize

- (Dictionary) To bring or restore to a normal condition

All entries sum to ONE

- Procedure:

- Step 1: Compute $Z = \text{sum over all entries}$
- Step 2: Divide every entry by Z

- Example 1

W	P
sun	0.2
rain	0.3

Normalize
→
 $Z = 0.5$

W	P
sun	0.4
rain	0.6

- Example 2

T	W	P
hot	sun	20
hot	rain	5
cold	sun	10
cold	rain	15

Normalize
→
 $Z = 50$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Probabilistic Inference

- Probabilistic inference: compute a desired probability from other known probabilities (e.g. conditional from joint)
- We generally compute conditional probabilities
 - $P(\text{on time} \mid \text{no reported accidents}) = 0.90$
 - These represent the agent's *beliefs* given the evidence
- Probabilities change with new evidence:
 - $P(\text{on time} \mid \text{no accidents, 5 a.m.}) = 0.95$
 - $P(\text{on time} \mid \text{no accidents, 5 a.m., raining}) = 0.80$
 - Observing new evidence causes *beliefs to be updated*



Inference by Enumeration

- $P(W)$?

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Inference by Enumeration

- $P(W)$?

$$P(\text{sun}) = .3 + .1 + .1 + .15 = .65$$

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Inference by Enumeration

- $P(W)$?

$$P(\text{sun}) = .3 + .1 + .1 + .15 = .65$$

$$P(\text{rain}) = 1 - .65 = .35$$

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Inference by Enumeration

- $P(W \mid \text{winter, hot})?$

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Inference by Enumeration

- $P(W \mid \text{winter, hot})?$

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Inference by Enumeration

- $P(W \mid \text{winter, hot})?$

$P(\text{sun} \mid \text{winter, hot}) \sim .1$
 $P(\text{rain} \mid \text{winter, hot}) \sim .05$

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Inference by Enumeration

- $P(W \mid \text{winter, hot})?$

$P(\text{sun} \mid \text{winter, hot}) \sim .1$
 $P(\text{rain} \mid \text{winter, hot}) \sim .05$
 $P(\text{sun} \mid \text{winter, hot}) = 2/3$
 $P(\text{rain} \mid \text{winter, hot}) = 1/3$

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Inference by Enumeration

- $P(W \mid \text{winter})?$

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Inference by Enumeration

- $P(W \mid \text{winter})?$

$$P(\text{sun}|\text{winter}) \sim .1 + .15 = .25$$

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Inference by Enumeration

- $P(W \mid \text{winter})?$

$$P(\text{rain} \mid \text{winter}) \sim .05 + .2 = .25$$

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Inference by Enumeration

- $P(W \mid \text{winter})?$

$P(\text{sun}|\text{winter}) \sim .25$

$P(\text{rain}|\text{winter}) \sim .25$

$P(\text{sun}|\text{winter}) = .5$

$P(\text{rain}|\text{winter}) = .5$

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Inference by Enumeration

- General case:

- Evidence variables: $E_1 \dots E_k = e_1 \dots e_k$
 - Query* variable: Q
 - Hidden variables: $H_1 \dots H_r$
- $$\left. \begin{array}{l} E_1 \dots E_k = e_1 \dots e_k \\ Q \\ H_1 \dots H_r \end{array} \right\} \begin{array}{l} X_1, X_2, \dots X_n \\ \text{All variables} \end{array}$$

- We want:

** Works fine with multiple query variables, too*

$$P(Q|e_1 \dots e_k)$$

- Step 1: Select the entries consistent with the evidence

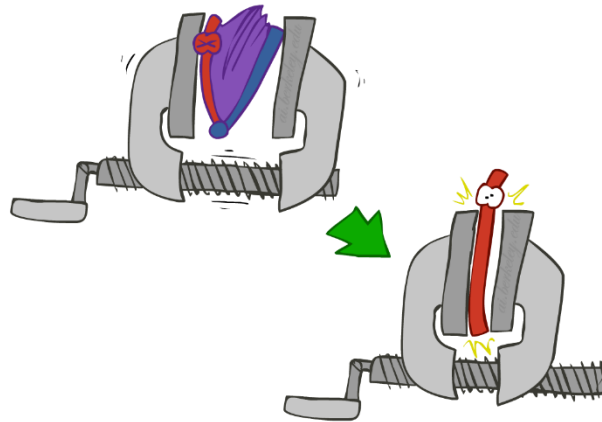


x	P(x)
-3	0.05
-1	0.25
0	0.07
1	0.2
5	0.01

2

0.15

- Step 2: Sum out H to get joint of Query and evidence



- Step 3: Normalize

$$\times \frac{1}{Z}$$

$$Z = \sum_q P(Q, e_1 \dots e_k)$$

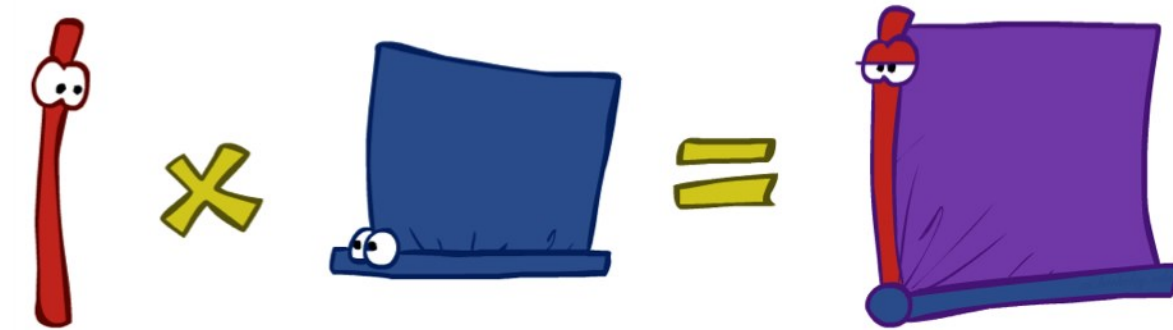
$$P(Q|e_1 \dots e_k) = \frac{1}{Z} P(Q, e_1 \dots e_k)$$

$$P(Q, e_1 \dots e_k) = \sum_{h_1 \dots h_r} \underbrace{P(Q, h_1 \dots h_r, e_1 \dots e_k)}_{X_1, X_2, \dots X_n}$$

The Product Rule

- Sometimes have conditional distributions but want the joint

$$P(y)P(x|y) = P(x, y) \quad \longleftrightarrow \quad P(x|y) = \frac{P(x, y)}{P(y)}$$



The Product Rule

$$P(y)P(x|y) = P(x, y)$$

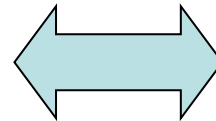
- Example:

$P(W)$

R	P
sun	0.8
rain	0.2

$P(D|W)$

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3



$P(D, W)$

D	W	P
wet	sun	
dry	sun	
wet	rain	
dry	rain	

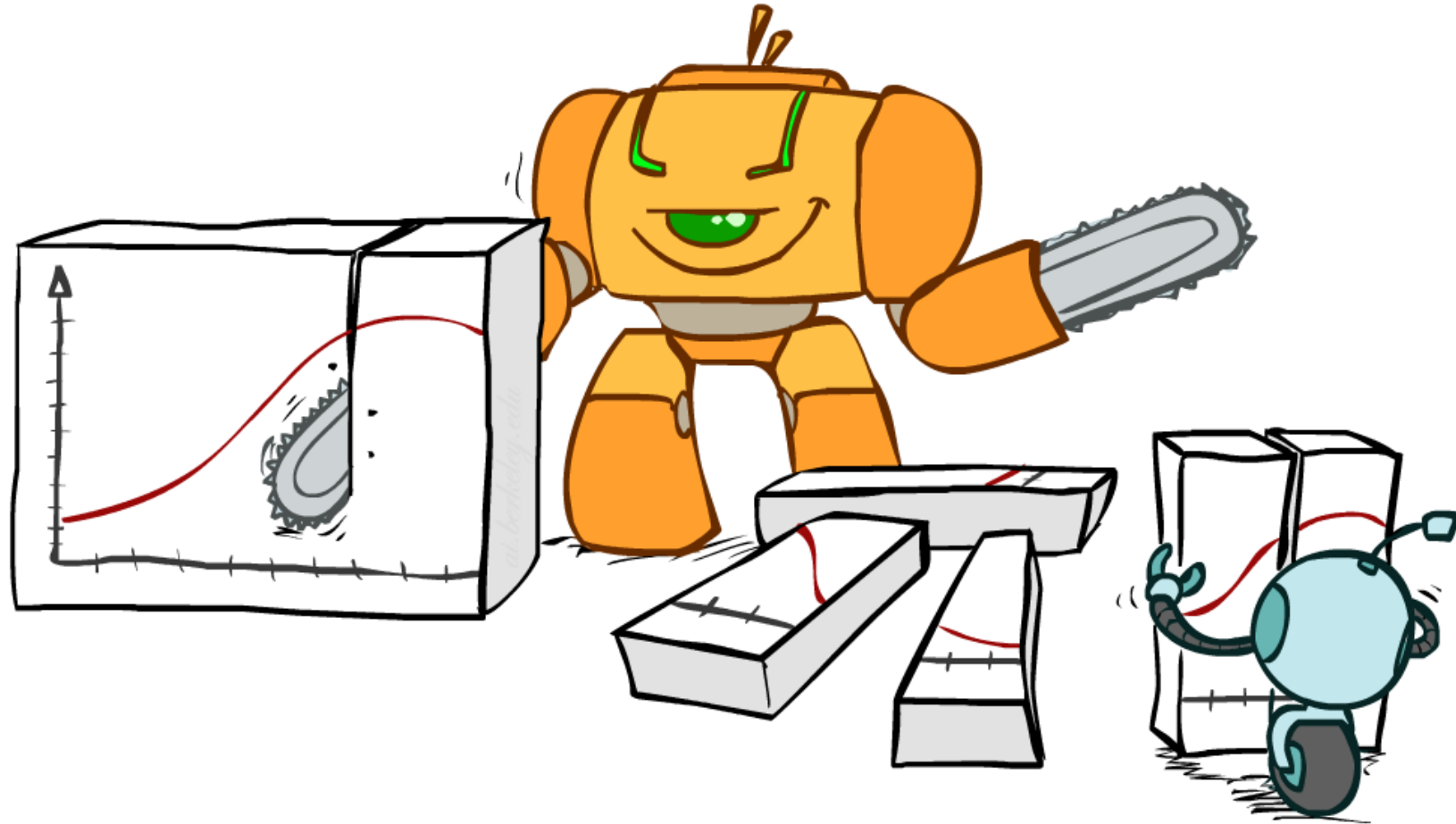
The Chain Rule

- More generally, can always write any joint distribution as an incremental product of conditional distributions

$$P(x_1, x_2, x_3) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2)$$

$$P(x_1, x_2, \dots, x_n) = \prod_i P(x_i|x_1 \dots x_{i-1})$$

Bayes Rule



Bayes' Rule

- Two ways to factor a joint distribution over two variables:

$$P(x, y) = P(x|y)P(y) = P(y|x)P(x)$$

- Dividing, we get:

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

- Why is this at all helpful?
 - Lets us build one conditional from its reverse
 - Often one conditional is tricky but the other one is simple
- In the running for most important AI equation!

That's my rule!



Inference with Bayes' Rule

- Example: Diagnostic probability from causal probability:

posterior
probability

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

Prior
probability

Effect
probability

- Example:

- M: meningitis, S: stiff neck

$$\left. \begin{aligned} P(+m) &= 0.0001 \\ P(+s|+m) &= 0.8 \\ P(+s|-m) &= 0.01 \end{aligned} \right\} \text{Example gives}$$

$$P(+m|+s) = \frac{P(+s|+m)P(+m)}{P(+s)} = \frac{P(+s|+m)P(+m)}{P(+s|+m)P(+m) + P(+s|-m)P(-m)} = \frac{0.8 \times 0.0001}{0.8 \times 0.0001 + 0.01 \times 0.999}$$

- Note: posterior probability of meningitis still very small
- Note: you should still get stiff necks checked out! Why?

Quiz: Bayes' Rule

- Given:

Prior
probability

$P(W)$

R	P
sun	0.8
rain	0.2

$P(D|W)$

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3

- What is $P(W \mid \text{dry})$?

Quiz: Bayes' Rule

- Given:

Prior probability

$P(W)$

R	P
sun	0.8
rain	0.2

$P(D|W)$

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3

- What is $P(W \mid \text{dry})$?

$$P(\text{sun}|\text{dry}) \sim P(\text{dry}|\text{sun})P(\text{sun}) = .9 \cdot .8 = .72$$

$$P(\text{rain}|\text{dry}) \sim P(\text{dry}|\text{rain})P(\text{rain}) = .3 \cdot .2 = .06$$

$$P(\text{sun}|\text{dry}) = 12/13$$

$$P(\text{rain}|\text{dry}) = 1/13$$

Probabilistic Graphical Model (PGM)

- It is a rich framework for encoding probability distributions over complex domains: joint (multivariate) distributions over large numbers of random variables that interact with each other.
- These representations sit at the intersection of statistics and computer science, relying on concepts from probability theory, graph algorithms, machine learning, and more.
- They are the basis for the state-of-the-art methods in a wide variety of applications, such as medical diagnosis, image understanding, speech recognition, natural language processing, and many, many more. They are also a foundational tool in formulating many machine learning problems.

Bayesian Networks

- A Bayesian network (also known as a Bayes network, Bayes net, belief network, or decision network) is a probabilistic graphical model that represents a set of variables and their conditional dependencies via a directed acyclic graph (DAG).
- Bayesian networks are ideal for taking an event that occurred and predicting the likelihood that any one of several possible known causes was the contributing factor.
- For example, a Bayesian network could represent the probabilistic relationships between diseases and symptoms. Given symptoms, the network can be used to compute the probabilities of the presence of various diseases.

Bayesian Networks

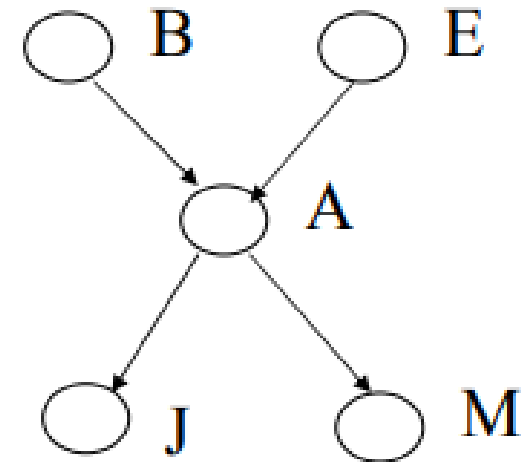
Two components: $B = (S, \Theta_S)$

- **Directed acyclic graph**
 - Nodes correspond to random variables
 - (Missing) links encode independences
- **Parameters**
 - Local conditional probability distributions for every variable-parent configuration

$$\mathbf{P}(X_i \mid pa(X_i))$$

Where:

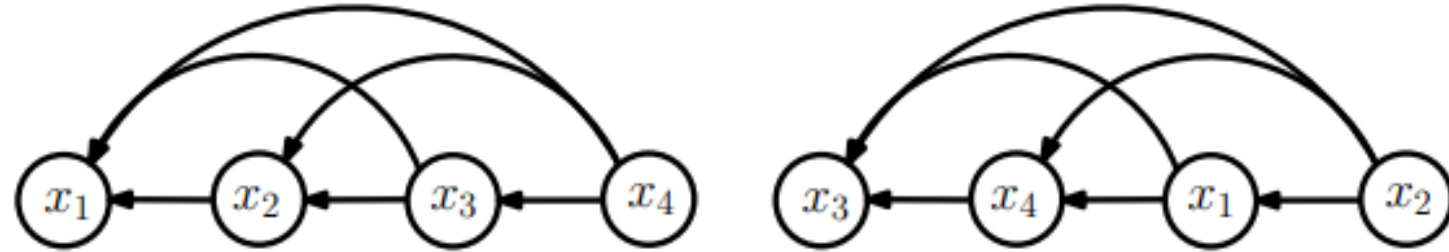
$pa(X_i)$ - stand for parents of X_i



$\mathbf{P}(A B,E)$			
B	E	T	F
T	T	0.95	0.05
T	F	0.94	0.06
F	T	0.29	0.71
F	F	0.001	0.999

Bayesian Networks

- Different Factorizations



- Two factorizations of four variables:

$$p(x_1, x_2, x_3, x_4) = p(x_1 \mid x_2, x_3, x_4)p(x_2 \mid x_3, x_4)p(x_3 \mid x_4)p(x_4)$$

$$p(x_1, x_2, x_3, x_4) = p(x_3 \mid x_1, x_2, x_4)p(x_4 \mid x_1, x_2)p(x_1 \mid x_2)p(x_2)$$

- Any distribution can be written in such a cascade form as a belief network (using chain rule)
- With independence assumptions the factorization often becomes simpler

Markov Model

- A Markov model is a stochastic model used to model pseudo-randomly changing systems.
- It is assumed that **future states depend only on the current state**, not on the events that occurred before it (that is, it assumes the Markov property).
- Generally, this assumption enables reasoning and computation with the model that would otherwise be intractable.
- For this reason, in the fields of predictive modelling and probabilistic forecasting, it is desirable for a given model to exhibit the Markov property.

Markov Model

- Markov chains are used for several business applications, including:
 - predicting customer brand switching for marketing
 - predicting how long people will remain in their jobs
 - predicting time to failure of a machine in manufacturing
 - forecasting the future price of a stock in finance
- Markov analysis is also used in natural language processing (NLP) and in machine learning.
 - For NLP, a Markov chain can be used to generate a sequence of words that form a complete sentence.
 - For machine learning, Markov decision processes are used to represent reward in reinforcement learning.

Markov Model

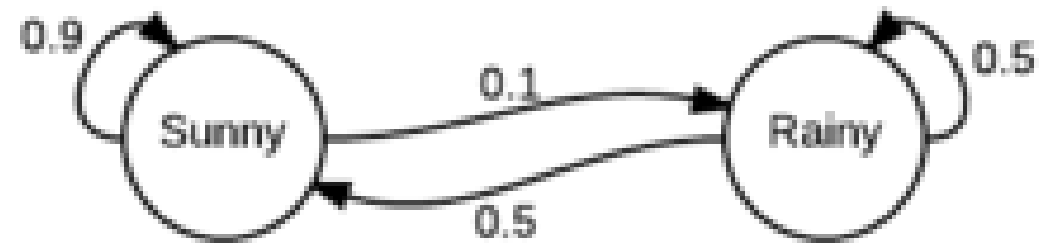
The weather on day 0 (today) is known to be sunny. This is represented by an initial state vector in which the "sunny" entry is 100%, and the "rainy" entry is 0%:

$$\mathbf{x}^{(0)} = [1 \quad 0]$$

The weather on day 1 (tomorrow) can be predicted by multiplying the state vector from day 0 by the transition matrix:

$$\mathbf{x}^{(1)} = \mathbf{x}^{(0)} P = [1 \quad 0] \begin{bmatrix} 0.9 & 0.1 \\ 0.5 & 0.5 \end{bmatrix} = [0.9 \quad 0.1]$$

Thus, there is a 90% chance that day 1 will also be sunny. The weather on day 2 (the day after tomorrow) can be predicted in the same way, from the state vector we computed for day 1:



$$\mathbf{x}^{(2)} = \mathbf{x}^{(1)} P = \mathbf{x}^{(0)} P^2 = [1 \quad 0] \begin{bmatrix} 0.9 & 0.1 \\ 0.5 & 0.5 \end{bmatrix}^2 = [0.86 \quad 0.14]$$

or

$$\mathbf{x}^{(2)} = \mathbf{x}^{(1)} P = [0.9 \quad 0.1] \begin{bmatrix} 0.9 & 0.1 \\ 0.5 & 0.5 \end{bmatrix} = [0.86 \quad 0.14]$$

Steady state of the weather $\mathbf{q} = \lim_{n \rightarrow \infty} \mathbf{x}^{(n)}$